Zer0pa / ZPE-IMC

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ZPE-IMC V0.0: MULTIMODAL MULTI-SENSORIAL CODEC: Text | Emojis | Images | Diagrams | Voice | Music | Touch | Taste | Smell | Mental

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Zer0pa-Architect-Prime ci: add auto-add-to-project workflow dc2a278 · 2 days ago

.github	ci: add auto-add-to-project workflow	2 days ago
code	Restore cold-clone public audit hotfixes	2 weeks ago
docs	fix change date from three years to five ye...	2 weeks ago
executable	Reconcile public authority surface	2 weeks ago
proofs	Restore ZPE-Test public packet on ZPE-IMC	2 weeks ago
.gitignore	wave4: public audit snapshot sync	2 weeks ago
.pre-commit-config.yaml	wave4: public audit snapshot sync	2 weeks ago
AUDITOR_PLAYBOOK.md	Restore ZPE-Test public packet on ZPE-IMC	2 weeks ago
CHANGELOG.md	Reconcile public authority surface	2 weeks ago
CITATION.cff	wave4: public audit snapshot sync	2 weeks ago
CODE_OF_CONDUCT.md	wave4: public audit snapshot sync	2 weeks ago
CONTRIBUTING.md	Restore ZPE-Test public packet on ZPE-IMC	2 weeks ago
GOVERNANCE.md	wave4: public audit snapshot sync	2 weeks ago
LICENSE	fix change date from three years to five ye...	2 weeks ago
Makefile	wave4: refresh public audit snapshot	2 weeks ago
PUBLIC_AUDIT_LIMITS.md	Restore cold-clone public audit hotfixes	2 weeks ago
README.md	Remove header from README	2 weeks ago
RELEASING.md	wave4: public audit snapshot sync	2 weeks ago
ROADMAP.md	wave4: public audit snapshot sync	2 weeks ago
SECURITY.md	Reconcile public authority surface	2 weeks ago

ZPE-IMC

Zero-Point Encoding

Integrated Multimodal Multisensory Codec

ZER0PA

license SAL v6.0 | python 3.11 | 3.12 | current authority 2026-03-07 accepted run | rust-backed accepted path | run-of-record 276.8k IMC words/s

quick verify setup & check | proof anchors manifest + run log | architecture runtime map | lane boundaries smell / taste caveats | deterministic byte-identical replay | cpu-native no GPU required

Quickstart & License	What This Is	Current Authority	Runtime Proof
Modality Snapshot	Throughput	Public ML Workbooks	Go Next

Quickstart And License

Quick Verify

The steps below are repository verification guidance, not packaged public-release instructions.

```
git clone https://github.com/Zer0pa/ZPE-IMC.git zpe-imc
cd zpe-imc
python -m venv .venv
source .venv/bin/activate
python -m pip install -e "./code[full,bench,dev]"
./code/rust/imc_kernel/build_install.sh
python - <<'PY'
from zpe_multimodal.core.imc import get_kernel_backend_info
print(get_kernel_backend_info())
PY
python ./executable/run_with_comet.py
# optional internal live logging only:
# COMET_API_KEY=... OPIK_API_KEY=... python ./executable/run_with_comet.py --enable-classic-comet --enable-opik
```

After a successful run you should see:

- `backend='rust'`
- `compiled_extension=True`
- `fallback_used=False`
- `python -m pytest code -q` is a valid direct public replay path after the editable install and uses the canonical `zpe_multimodal.*` package surface
- Local rerun bundle emitted under `proofs/reruns/IMC-Canonical-<UTC timestamp>/`
- Shipped operator reference artifacts remain at `proofs/logs/phase6_run_of_record_manifest.json` and `proofs/logs/phase6_comet_run.txt`

Quickest outsider orientation:

docs/FAQ.md	docs/SUPPORT.md	docs/ARCHITECTURE.md

License Boundary

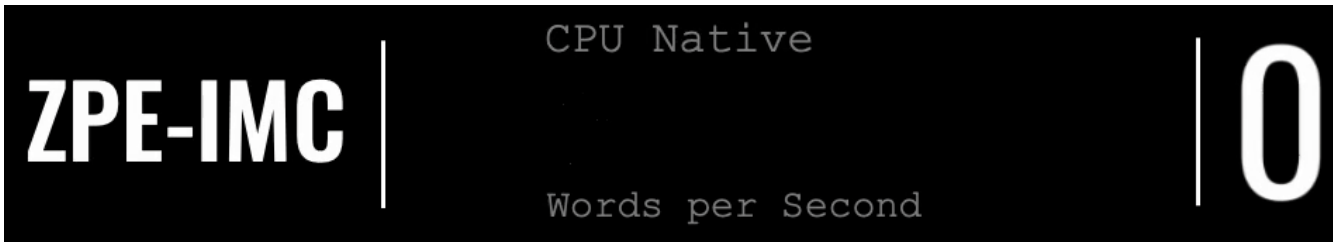
- Free to use at or below USD 100M annual gross revenue under SAL v6.0.
- SPDX tag: `LicenseRef-Zer0pa-SAL-6.0`.
- Commercial or hosted use above that threshold requires contact at architects@zer0pa.ai.
- Release chronology lives in `CHANGELOG.md` and `CITATION.cff` — it is not clone/install guidance.

What This Is

ZPE-IMC is the Wave-1 integration and dispatch layer for Zero-Point Encoding. It is a deterministic, CPU-native transport system that carries ten user-facing modalities through a single 20-bit transport word — no neural network, no training loop, and no GPU. Those ten modalities are routed through eight lane families: `TEXT_EMOJI` , `DIAGRAM_IMAGE` , `MUSIC` , `VOICE` , `MENTAL` , `TOUCH` , `SMELL` , and `TASTE` .

The core claim is transport integrity across mixed modalities. This README is the current authority snapshot ahead of the first tagged public release. Use it for first-contact comprehension, verification, and routing into the proof corpus.

Question	Answer
What is this?	A deterministic, unified multimodal transport system with real modality integration — not a tokenizer-first framing, and not a set of separate codec demos.
What is the current authority state?	The March 7, 2026 Rust-backed saturated run is the accepted operator authority: `IMC-Canonical-20260307T131330Z` with `170 passed` in the full operator lane and `benchmark_run_id=A4-BENCH-20260307T131414Z` ; the public snapshot rerun truth is `169 passed, 1 skipped` .
What is actually proved?	Four things: native backend truth (`backend=rust` , `compiled_extension=1` , `fallback_used=0`), deterministic byte-identical replay, integrated coverage across all ten user-facing modalities, and current throughput authority of `276798.7185 imc_stream_words/sec` .
What is not being claimed?	No claim of human-equivalence semantics, commodity compression supremacy, unconstrained smell/taste coverage, or phoneme-perfect / speaker-ID voice equivalence.
Where should an outsider acquire and verify?	Clone from `https://github.com/Zer0pa/ZPE-IMC.git` and run `python ./executable/run_with_comet.py` . Inspect the rerun bundle under `proofs/reruns/IMC-Canonical-<UTC timestamp>/` . The shipped `proofs/logs/phase6_*` pair serves as the stable operator reference.



Current Authority

Accepted run-of-record	Backend truth	Current throughput authority
`IMC-Canonical-20260307T131330Z` [Comet] Later March 7 Rust-backed saturated run; supersedes the earlier same-day rerun.	`backend=rust` , `compiled_extension=1` , `fallback_used=0` Authority path is the compiled native extension, not a Python fallback.	`canonical_total_words_per_sec=276798.7185` , `throughput_encode_words_per_sec=94104.7837` , `throughput_decode_words_per_sec=296145.6735` [Comet] These are the accepted front-door headline numbers for the Rust-backed path.

Surface	Locked value	Why it matters
Accepted run-of-record	`IMC-Canonical-20260307T131330Z` [Comet]	Later March 7 Rust-backed saturated run; supersedes the earlier same-day rerun.
Test state	`operator: 170 passed` ; `public snapshot: 169 passed, 1 skipped`	The public lane skips one operator-only Triton byte-identity check when the private A6 export is absent. This is intentional.

Surface	Locked value	Why it matters
Backend truth	`backend=rust` , `compiled_extension=1` , `fallback_used=0`	Authority path is the compiled native extension, not a Python fallback.
Current throughput authority	`canonical_total_words_per_sec=276798.7185` , `throughput_encode_words_per_sec=94104.7837` , `throughput_decode_words_per_sec=296145.6735` [Comet]	These are the accepted front-door headline numbers for the Rust-backed path.
Throughput unit	`imc_stream_words/sec`	All README throughput numbers use transport words, not natural-language words per second.
Authority artifacts	`proofs/logs/phase6_run_of_record_manifest.json` , `proofs/logs/phase6_comet_run.txt`	These are the current proof anchors for runtime truth.
External acquisition surface	`https://github.com/Zer0pa/ZPE-IMC.git`	Outsider clone target. May lag the source repo.

Three Dimensions Of Authority

Integrated Transport The primary authority surface. The system carries all ten modalities through a single deterministic transport layer.	**Rust Runtime** Subordinate to transport but real. The compiled Rust backend preserves semantics, determinism, proofs, and observability while exceeding the Python baseline.	**Provenance** Separate from runtime authority. Custody and lineage records explain how the current state emerged; archive material is never promoted to live operator truth.

Authority Notes

The source repository is the single truth surface. Uploaded snapshots may omit files or carry path defects — they never outrank the source repo.	The `total_words=844` Wave-1 demo anchor is frozen for compatibility and historical reference. It is not the authority path for the live Rust-backed kernel.	Historical and archive material explains lineage and caveats but does not replace the manifest/log pair as runtime authority.

Runtime Proof (Wave-1)

All locked runtime values are in the Current Authority table above. To verify locally: build from `code/rust/imc_kernel/build_install.sh` , confirm the backend with `zpe_multimodal.core.imc.get_kernel_backend_info()` , and treat the manifest/log pair as the top proof surfac

Run-of-record manifest `PASS` Live authority artifact carrying backend truth, saturation facts, benchmark identity, and proof URLs.	**Native backend truth** `backend=rust` , `compiled_extension=1` , `fallback_used=0` The accepted runtime is the compiled Rust extension, not a Python fallback.
Deterministic replay `determinism_hash_match=1` , `all_deterministic=1` [Comet]	**Modality coverage** `modality_coverage_count=10` , `modality_coverage_all=1` [Comet]

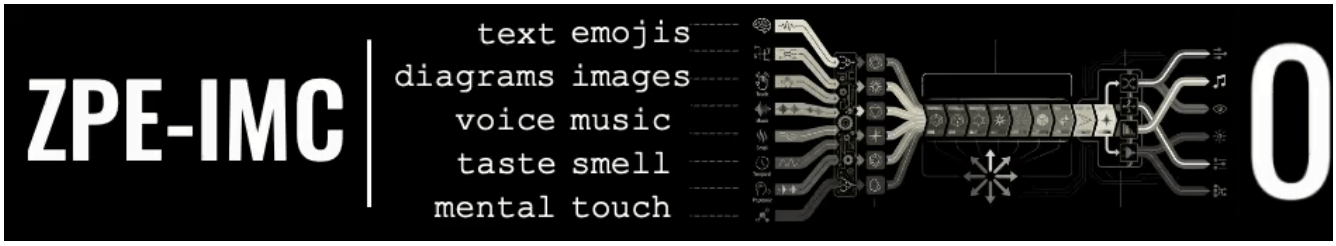
README Code of conduct Contributing License Security

Proof Anchors

`proofs/logs/phase6_run_of_record_manifest.json` backend truth, saturation facts, benchmark identity, and live proof links.	`proofs/logs/phase6_comet_run.txt` Accepted wrapper run log carrying the locked runtime values.
`code/benchmarks/artifacts/BENCHMARK_REPORT.md` Benchmark report referenced by both the manifest and the run log.	`docs/ARCHITECTURE.md` Runtime map and authority class definitions.

Proof rung	Locked value	What it proves now
Run-of-record manifest	`PASS`	Live authority artifact carrying backend truth, saturation facts, benchmark identity, and

Proof rung	Locked value	What it proves now
		proof URLs.
Native backend truth	`backend=rust` , `compiled_extension=1` , `fallback_used=0`	The accepted runtime is the compiled Rust extension, not a Python fallback.
Accepted March 7 rerun	`run_name=IMC-Canonical-20260307T131330Z` , `170 passed` in the full operator lane, `8` workers, `benchmark_run_id=A4-BENCH-20260307T131414Z`	This is the later accepted March 7 run-of-record and supersedes the earlier same-day rerun; the public snapshot rerun truth is the separate `169 passed, 1 skipped` lane.
Current throughput authority	`canonical_total_words_per_sec=276798.7185` , `throughput_encode_words_per_sec=94104.7837` , `throughput_decode_words_per_sec=296145.6735` [Comet]	Accepted saturated steady-state wrapper ceiling for the Rust-backed path.
Deterministic replay	`determinism_hash_match=1` , `all_deterministic=1` [Comet]	Replay is byte-identical on the accepted proof path.
Modality coverage	`modality_coverage_count=10` , `modality_coverage_all=1` [Comet]	The promoted path integrates all ten user-facing modalities.
Historical demo anchor	`844` Wave-1 demo	Frozen compatibility and historical context only; not the current runtime authority.



Modality Status Snapshot

Status is reported per modality across all ten user-facing modalities, even where the underlying lane families share folders and transport markers. Text and emoji, diagram and image, and music and voice each appear as separate rows below; their shared lane roots are unchanged.

Lane Boundaries

Music Deterministic transport with preserved `time_anchor_tick` . The multi-tempo limitation remains visible.	**Voice** Descriptor-aware deterministic transport. Not a claim of phoneme-perfect semantics, speaker-ID equivalence, or full speech understanding. Source-mode policy/code alignment remains an open caveat.
Smell Subset-bounded authority covering the active `SmellNet_HF` + `OpenPOM_GS_LF` subset — read as `READY_WITH_LICENSE_BOUNDARY` , not unconstrained olfaction.	**Taste** All existing caveats remain attached: `0x0400` overlap hygiene, derived-evidence history, portability lineage, and `ChemTastesDB` commercial exclusion.

Lane family	Modality	Status	Proved now	Boundary and evidence
`TEXT_EMOJI`	Text	`GREEN`	Long-form text transport is deterministic and reversible on the accepted March 7 path.	`300/300` external baseline round-trip valid; determinism `5/5` ; hop `6/6` ; `encoded_words=101,999` for `chars=94,999` .
`TEXT_EMOJI`	Emoji	`GREEN`	Exact round-trip is current authority, including ZWJ and skin-tone sequences in the shared lane family.	`4,973` handled; exact round-trip in canonical checks; deterministic replay retained.
`DIAGRAM_IMAGE`	Diagram	`GREEN`	Structure, inherited styles, and transforms are preserved with deterministic decode.	`16/16` pytest pass; mean path-distance `0.44–0.79` .

Lane family	Modality	Status	Proved now	Boundary and evidence
`DIAGRAM_IMAGE`	Image	`GREEN`	Mixed-stream image transport runs deterministically on the accepted Rust-backed path with native encode/decode.	Transport-integrity claim, not commodity image-compression supremacy; PSNR `45.95 dB` Earthrise, `39.10 dB` Mona Lisa; byte-identical replay across `5` runs.
`MUSIC`	Music	`GREEN`	Closure is current authority with preserved `time_anchor_tick`.	`7/7` fixtures pass; `events=4`; `packed_words=34`; multi-tempo limitation remains visible.
`VOICE`	Voice	`GREEN`	Descriptor-aware deterministic transport is integrated on the promoted path.	`4/4` fixtures pass; `parity_all_pass=true`; `determinism_all_same=true`; not a speech-understanding or speaker-equivalence claim.
`MENTAL`	Mental	`GREEN`	Spatial and cognitive structure encoding (D6-12 profile) passes the authority test suite.	`28/28` pytest pass; IMC parity gate pass; no mind-reading or clinical-equivalence claim.
`TOUCH`	Touch	`GREEN`	Haptic and proprioceptive transport is integrated on the authority path; compression and IMC parity evidence retained.	`20/20` pytest pass; IMC parity gate pass; raw `549` bytes to ZPE `87` bytes.
`SMELL`	Smell	`GREEN`	Authority is real but subset-bounded.	`116` comparator cases; active subset deterministic; public authority is `Q-103 = READY_WITH_LICENSE_BOUNDARY`.
`TASTE`	Taste	`GREEN`	Integrated on the current authority path.	`bitter=1,986`; `sweet=8,280`; `sour=1,505`; `umami=326`; `salty=58`; `6` anchor round-trip cases; keep all existing caveats attached.

Throughput

Throughput numbers, latency headlines, and benchmark identity are locked in the Current Authority table above. The accepted performance authority is the later saturated Rust-backed run recorded in the manifest/log pair. Older hardware comparison tables are historical benchmark ancestry — not the current operator ceiling.

Canonical throughput `276798.7185` [Comet] Accepted saturated steady-state parallel-batch transport throughput.	**Encode throughput** `94104.7837` [Comet] Accepted wrapper encode throughput on the native path.
Decode throughput `296145.6735` [Comet] Accepted wrapper decode throughput on the native path.	**Short-text latency p50** `0.377 ms` [Comet] Current accepted short-text benchmark headline from the run-of-record manifest.

Measure	Locked value	Meaning
Rate unit	`imc_stream_words/sec`	All throughput figures below are transport words, not natural-language words per second.
Run name	`IMC-Canonical-20260307T131330Z` [Comet]	Later accepted March 7 run-of-record.
Benchmark run id	`A4-BENCH-20260307T131414Z` [Comet]	Current benchmark identity mirrored by the manifest, run log, and benchmark artifacts.
Canonical throughput	`276798.7185` [Comet]	Accepted saturated steady-state parallel-batch transport throughput.
Encode throughput	`94104.7837` [Comet]	Accepted wrapper encode throughput on the native path.

Measure	Locked value	Meaning
Decode throughput	`296145.6735` [Comet]	Accepted wrapper decode throughput on the native path.
Short-text latency p50	`0.377 ms` [Comet]	Current accepted short-text benchmark headline from the run-of-record manifest.

Public ML Workbooks

These are the public Comet workbooks that map cleanly to the live README claims. The first row is the current public twin for the cited ru identity and headline metrics; the remaining rows are useful lineage only.

Role	Run name	Benchmark run id	Workbook
Current promoted public twin	`IMC-Canonical-20260307T131330Z`	`A4-BENCH-20260307T131414Z`	`1d08a49dfc2e4684bc3f537117df981d`
Historical lineage	`IMC-Canonical-20260307T124656Z`	`A4-BENCH-20260307T124746Z`	`1e54c33b10aa458d8a09d485d38abfbc`
Historical lineage	`IMC-Canonical-20260307T113918Z`	`A4-BENCH-20260307T114002Z`	`5458458688344f52bb85128e3d71cda9`
Historical lineage	`IMC-Canonical-20260307T102831Z`	`A4-BENCH-20260307T102931Z`	`7f8d428d065a446aba03518e654b06ce`
Historical lineage	`IMC-Canonical-20260307T012457Z`	`A4-BENCH-20260307T012849Z`	`e13cbb4c8bd34130923d65c150200591`

Go Next

If you need to…	Open this
Understand the runtime map and authority classes	`docs/ARCHITECTURE.md`
Install, verify, and answer common first-contact questions	`docs/FAQ.md`
Read legal and lane-specific public boundaries	`docs/LEGAL_BOUNDARIES.md`
Audit historical compatibility and replay boundaries	`AUDITOR_PLAYBOOK.md`
Read public audit limits and explicit non-claims	`PUBLIC_AUDIT_LIMITS.md`
Inspect the benchmark report behind the accepted run	`code/benchmarks/artifacts/BENCHMARK_REPORT.md`
Inspect proof artifacts and logs directly	`proofs/`

Area	Purpose
`README.md` , `CHANGELOG.md` , `CONTRIBUTING.md` , `SECURITY.md` , `CODE_OF_CONDUCT.md` , `CITATION.cff` , `LICENSE`	Root governance and release-facing metadata
`code/`	Installable package and codec implementation surface
`docs/`	Interface contracts, FAQ, support, and lane documentation

Area	Purpose
`proofs/`	Proof corpus, baselines, and falsification evidence
`executable/`	Executable runtime authority path
`(external ops archive)`	Operational wave tracking artifacts are archived outside this repository to keep the upload surface lean

Open Risks (Non-Blocking)

- The optional audio dependency chain may fail on Python 3.14. Python 3.11 and 3.12 remain the practical baseline for full audio paths.
- The public repo at `https://github.com/Zer0pa/ZPE-IMC.git` can lag the live working tree. Within any acquired tree, use the manifest/log pair and current docs as the authority root.
- Some scripts and docs still contain machine-absolute paths and need portability cleanup.
- Live cloud reruns require valid `COMET_API_KEY` and `OPIK_API_KEY` in the operator environment.
- H200 validation is deferred pending replay on actual H200 hardware under the locked `WS3` protocol. Do not publish H200 comparati performance claims until that replay evidence exists.

Contributing, Security, Support

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 2

Zer0pa-Architect-Prime
Copilot

Languages

Python 90.6% Rust 6.9% JavaScript 1.6% Other 0.9%